Richard B. Anderson

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Education

The University of Texas at El Paso (UTEP) Anticipated Graduation : May 2026
Bachelor of Science in Physics
Bachelor of Science in Mathematics Overall GPA: 3.74/4.00

Research Experience

Undergraduate Research Intern - The University of Texas at Arlington
The University of Texas at Arlington Neutrino and Rare Event Searches (NuRES) Group *May 2024 - Present*

- Conducted research on neutrinoless double beta decay as part of the NEXT experiment.
- Analyzed S1 and S2 signals to improve detector sensitivity.
- Simulated particle events using the Geant-4 Nexus framework to study alpha-induced background radiation.
- Automated waveform analysis across large datasets, extracting key parameters like signal decay constants and peak widths.
- Developed visualizations for detector event data and contributed to the understanding of xenon gas ionization patterns.
- Applied effective field theory techniques to explore beyond Standard Model physics in double beta decay.
- Used nuDoBE Python tool to calculate decay rates and electron kinematics under different Wilson coefficients.
- Analyzed electron energy and angular spectra to compare between Majorana neutrino models and other lepton number violation contributions.

Undergraduate Research Assistant
UTEP - Astrophysics Theory Group *January 2023 - Present*

- Conducting *Astrophysical* research under Dr. Kedron Silsbee, studying the capture rate of planetesimals by collapsing proto-stars.
- Utilized Slurm *High-Performance Computing* cluster to run Python scripts to simulate physical phenomena and analyze subsequent data to come to a physically significant conclusion.
- Utilized knowledge of classical mechanics and mathematics to create analytical models of celestial bodies, as well as statistical predictions of particle motion.
- Presented preliminary findings at the Spring COURI Symposium poster session.

Undergraduate Research Assistant - UTEP Aerospace Center
UTEP Aerospace Center - Missile Systems Innovation Team *September 2023 - March 2024*

- Effectively employed *Python* machine learning and computer vision packages, such as *OpenCV* and *TensorFlow*, to implement neural networks for object detection and tracking.
- Utilized drone hardware and *ArduPilot* software for object tracking.
- Experience collaborating with a multidisciplinary team dedicated to utilizing digital engineering techniques to enhance and innovate missile systems, with a focus on achieving *cost-effective solutions*.
- Applied numerical and analytical techniques to calculate ballistic trajectories and optimize missile sizing and design parameters.
- Completed training on basics and fundamentals of Siemens NX software.
- Leveraged digital development environments to thoroughly assess the accuracy and efficacy of object tracking scripts.

Teaching Experience

Academic Tutor
El Paso Community College - Student Success *July 2022 - July 2023*

- Assisted students of El Paso Community College with mathematics coursework in courses ranging from remedial mathematics courses to differential equations.
- Coordinated with other tutoring team members during high-traffic hours to efficiently and effectively address the high volume of students.
- Utilized both in person tutoring sessions, and navigated online learning interface to deliver online tutoring option for students.
- Proficiently utilized Microsoft Suite programs for seamless communication within the tutoring team and to meticulously track student data.

Awards & Activities

Terry Scholarship
The Terry Foundation *Fall 2023 – Spring 2027*

- Awarded a prestigious academic scholarship of $25,000 per academic year, based on high academic achievement and leadership potential.
- Engage in up to 15 hours per semester of community service activities in the El Paso community.

COURI SURPASS Program
The University of Texas at El Paso *June 2023 – August 2023*

- Participated in a Summer Research Experience for Undergraduates with a financial award of up to $2,000 for the summer.
- Attended academic research workshops and professional development training relevant to scientific research.
- Conducted research under Dr. Kedron Silsbee, contributing to research in planet formation.

Nuclear Physics Summer School for Underrepresented Students (NuPEERS)
Dillard University, New Orleans, LA *June 2023*

- Participated in a competitive summer school program focused on nuclear physics, supported by the Department of Energy's Office of Nuclear Physics.
- Attended lectures and hands-on workshops covering advanced topics in nuclear physics, including neutrino physics, machine learning, and surveys of modern nuclear physics research.
- Collaborated with peers from Minority Serving Institutions (MSIs) to solve challenging problems and explore innovative research in nuclear science.
- Received a financial award of $500 for successful completion of the program.

Leadership Experience

UTEP Astronomy Club - Secretary

- Founding member and officer of the UTEP astronomy club.
- Playing a key role in organizing and hosting observational astronomy events, enhancing public engagement with astronomy through educational presentations and night sky viewings.

Technical Skills

C++ | Python | Fortran | Bash Scripting | Mathematica | Git | SLURM HPC | HTML | LaTex |